

# Improved Euler's Method

## Key Points

Consider the differential equation

$$\frac{dy}{dx} = f(x, y)$$

For a small value  $h$ , let  $x_1 = x_0 + h$ .

Set  $y_1^* = y_0 + h f(x_0, y_0)$ . An even better approximation of  $y(x_1)$  is

$$y_1 = y_0 + \frac{h}{2} (f(x_0, y_0) + f(x_1, y_1^*))$$

Repeating this inductively, we obtain

$$y_{n+1} = y_n + \frac{h}{2} (f(x_n, y_n) + f(x_{n+1}, y_{n+1}^*))$$

where  $x_{n+1} = x_n + h$ , and

$$y_{n+1}^* = y_n + h f(x_n, y_n)$$

## Basic Skills

**Q1.** If  $x_0 = 5$  and  $h = 0.2$ , then calculate

- a)  $x_1$    b)  $x_5$    c)  $x_7$    d)  $x_{10}$

**Q2.** Let  $f(x, y) = x^2 + xy$ . Calculate  $f(x_0, y_0)$  if

- a)  $(x_0, y_0) = (1, 1)$    b)  $(x_0, y_0) = (2, 3)$   
c)  $(x_0, y_0) = (5, 2)$    d)  $(x_0, y_0) = (7, -4)$

**Q3.** If  $f(x, y) = \ln(x + y)$  and  $h = 0.25$ , then calculate  $y_1^*$  when

- a)  $(x_0, y_0) = (0, 1)$    b)  $(x_0, y_0) = (3, 1)$

**Q4.** If  $f(x, y) = x^2 + y^2$  and  $x_0 = 2$ ,  $y_0 = 5$ ,  $h = 0.25$  and someone calculated Euler's step estimate  $y_1^* = 12.25$  then calculate Euler's improved step estimate  $y_1$

## Competency Skills

**Q5.** Consider the differential equation  $\frac{dy}{dx} = xy + x$ . Let  $x_0 = 0$ ,  $y_0 = y(0) = 1$ , and  $h = 0.25$ .

- a) Calculate  $y_1^*$   
b) Use this to calculate  $y_1$   
c) Fill in the following table  
d) Use the table to state the estimate for  $y(1)$

| $i$   | 0 | 1 | 2 | 3 | 4 |
|-------|---|---|---|---|---|
| $x_i$ |   |   |   |   |   |
| $y_i$ |   |   |   |   |   |

**Q6.** Consider the differential equation  $\frac{dy}{dx} = e^{x^2 y} + 3x + y$ . Let  $x_0 = 0$ ,  $y_0 = y(0) = 3$ , and  $h = 0.5$ . Use the improved Euler's method to approximate  $y(1)$

**Q7.** Consider the differential equation  $\frac{dy}{dx} = (x + y)^3$ . Let  $x_0 = 1$ ,  $y_0 = y(0) = 5$ , and  $h = 1$ . Use the improved Euler's method to approximate  $y(2)$  and explain why the Improved method gets a larger answer

## Advanced Skills

- Q8. a)** Solve the differential equation  $\frac{dy}{dx} = xy + x$   
b) Find  $y(1)$  and compare it to the answer found by Euler's improved step method in Q5  
c) Find  $y(1)$  by Euler's (non-improved) step method and compare it to Euler's improved step method.  
d) Decide for this differential equation if it is worth the effort to use Euler's improved step method over the standard Euler's step method.
- Q9.** Explain how Euler's improved step method improves the original method
- Q10.** Find  $y(1)$  given  $\frac{d^2 y}{dx^2} = 2 \frac{dy}{dx} - 3y + x$  and  $\frac{dy}{dx}(0) = 2$ ,  $y(0) = 0$  by using Euler's improved step method with  $h = 1$

## Extension

**Q11.** Consider the rate of decay of a radioactive substance given by  $\frac{dN}{dt} = -\lambda N_0 e^{-\lambda t}$ , where  $N(t)$  is the amount of the substance at time  $t$ ,  $N_0$  is the initial amount of the substance, and  $\lambda$  is the decay constant. Estimate the percentage of substance left at  $t = 3$  in terms of  $\lambda$ .